1. Allow the user to enter a quantity and price, use ctl+z to stop. Use a function to compute the total (quantity times price). The function should be passed the quantity and price and then return the total. Use another function to compute 10% discount if the total is over $10,0000.00 and 5% for any amount equal to or lower than $10,000.00. The second function should receive the total, check which discount rate to charge, compute discount amount and then compute the discount total (total – discount amount). It should return the discount total. Display total and
discount total. Sum total and discount total and display at the end.

Input	Process	Output

quantity	double computeTotal(int quantity, double price)
	{
	return quantity * price;
	}
		Total:

Price	double computeDiscountTotal(double total)
	
	{
	
	If/else conditions
	}
		Discount total:

total	While loop	Sum total: Sum Discount total:
discountTotal		
sumDiscountTotal		
sumTotal		

2. Enter players last name, number of hits and at bats at the keyboard, use ctl+z to stop. Use a function to compute batting average. Pass the hits and at bats to the function. The function should return batting average (at bats / number of hits). Display last name and batting average. Give a count of the number of players entered and display the count after the loop.

Input	Process	Output
Lname	Double batting_average(int hits, int atbats) { Return if/return statement	Last Name: Batting Average: Total Player Count:

	}	
Number of hits	While (last name >> hits >> at bats)	
At bats		
Average		
count		

3. Enter the destination city, miles travelled and gallons used for a trip, use ctl+z to stop. Use a function to compute miles per gallon. Pass miles travelled and gallons used to the function. The function should return miles per gallon. Use another function to compute gas cost. Pass to this function gallons used. Each gallon costs $3.50. Compute and return the cost. Display destination city, miles per gallon and cost of gas. Sum and display the total cost of gas.

Input	Process	Output

city	Double mpg(double gallons, double miles)
	{
	If/return statement)
	}
		Destination:

Miles	Double gasCost (double gallons , double galloncost)	MPG:

Gallons	While (getline(cin, city)
	
	{
	If/else condition with (!) operator
	}
		Cost of gas:

Cost per gallon		Total Cost of Gas:

4. Allow the employee to enter last name, job code and hours worked, use ctl+z to stop. Use a function to determine the pay rate. Pass to this function the job code and it should return rate of pay. Use the following rates based on Job code: L is $25/hr, A is $30/hr and J is $50/hr for respective pay rates. Write another function to determine the gross pay. Pass to this function the hours worked and pay rate and return gross pay. Give time and a half for overtime. Display last name and gross pay. Sum and display total of all gross pay.

Input	Process	Output

lname	Double payRate(char jobCode)
	{
	Switch Conditions
	}
		Last name:

rate	Double grossPay(int hours, char jobCode)
	{
	If/return condition for overtime pay
	
	}
		Gross pay:

Hours	While (lname >> jobCode >> hours) If (payrate == 0) Cout<< “Invalid response” Else Gross pay = grosspay(hours, jobCode)	Total gross pay:
Gross		
totalGross		

5. Allow the user to enter student last name, credit hours and district code, use ctl+z to stop. Use a function to compute tuition owed. First, write a function to determine the cost per credit hour. Charge In district (code of I) $250 per credit hour. Out of district (code of O) is $550 per credit hour. Write another function to compute tuition cost. Display student name and tuition cost. Sum and display total of all tuition costs.

Input	Process	Output
lname	Double cost per credit (char district) Switch (toupper (static_*cast<unsigned_char>(district)* *Case 'I' : return 250.00;* *Case 'O' : return 550.00;* *Default : 0.0*	Student Last Name:

credits	Double computeTuition(int credits , char district)
	{
	Return credits * costPerCredit(district)
	}
		Tuition Cost:

District	While (lname>>credits>>district)
	
	If (costPerCredit ==0)
	{
	Cout << " Invalid entry please enter I or O for district code"<< endl
	}
	Else
	{
	Tuition = computeTuition(credits* costPerCredit(district)
	
	Total_tuition+= tuition
	
	Cout<< Student name<<lname<< endl;
	Cout<< Tuition Cost<< tuition<< endl;
	}
	Cout<< Sum of total tuition<< total_tuition<< endl;
		Sum of Total Tuition:

tuition		
Total_tuition		
costPerCredit		
Compute_tuition		